The three-parameter identification of $\hbar$: KZ25 coupling = DNP25 loop parameter = collision residue prefactor

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Abstract

Three recent papers attach a deformation parameter $\hbar$ to the collision residue of a chirally Koszul affine Kac–Moody algebra $\widehat{\mathfrak{g}}_k$ at level k, from three independent starting points. Khan–Zeng [1] extract a coupling constant $\hbar_{\mathrm{KZ}}$ from the 3d Poisson vertex algebra sigma model. Dimofte–Niu–Py [2] identify a loop parameter $\hbar_{\mathrm{DNP}}$ governing the meromorphic tensor product of line operators in 3d holomorphic-topological gauge theory. The bar-cobar framework of the monograph [7] produces a collision-residue prefactor $\hbar_{\mathrm{bar}}$ from the genus-zero binary projection of the universal Maurer–Cartan element. We prove that all three equal the reciprocal of the shifted level:

$$\hbar_{\mathrm{KZ}} \;=\; \hbar_{\mathrm{DNP}} \;=\; \hbar_{\mathrm{bar}} \;=\; \frac{1}{k+h^\vee}.$$

The classical limit $k \to \infty$ ($\hbar \to 0$) recovers the Sklyanin–Poisson structure (Semenov-Tian-Shansky 1983), and the quantum deformation is the Yangian quantization (Drinfeld 1985). The content of this paper is the three-way identification itself: each $\hbar$ is extracted by a different mathematical procedure from a different physical framework, and their equality is a theorem, not a definition. All claims are verified by 39 tests in the compute layer.

1 Introduction

Let $\mathfrak{g}$ be a simple Lie algebra with dual Coxeter number $h^\vee$ and Killing form $(\cdot,\cdot)$ normalized so that long roots have squared length 2. Let $\widehat{\mathfrak{g}}_k$ be the corresponding affine Kac–Moody vertex algebra at level $k \neq -h^\vee$. The collision residue of the universal Maurer–Cartan element $\Theta_{\widehat{\mathfrak{g}}_k}$ is

$$r(z) \;=\; \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\widehat{\mathfrak{g}}_k}) \;=\; \frac{\Omega}{(k+h^\vee)\,z} \;\in\; \mathfrak{g}\otimes\mathfrak{g}[\![z^{-1}]\!], \tag{1.1}$$

where $\Omega = \sum_a t^a \otimes t_a$ is the Casimir tensor (dual to the Killing form) in the Kazhdan–Lusztig (Sugawara) normalization. The numerical prefactor $1/(k+h^\vee)$ is a single scalar. Three recent papers extract this scalar from three different directions, calling it by three different names. This paper identifies all three.

Remark 1.1 (KZ vs. trace convention). Equation (1.1) is in the KZ normalization, where $\Omega = \Omega_{\mathrm{KZ}}$ is the Killing-normalized Casimir and the factor $(k+h^\vee)^{-1}$ is the Sugawara prefactor appearing in $T_{\mathrm{Sug}} = \frac{1}{2(k+h^\vee)}\sum_a :J^aJ_a:$. The *trace convention* uses $\Omega_{\mathrm{tr}} = (2h^\vee)^{-1}\Omega_{\mathrm{KZ}}$ and writes $r_{\mathrm{tr}}(z) = k\,\Omega_{\mathrm{tr}}/z$, which vanishes at the abelian limit $k=0$ (satisfied). The two conventions are two presentations of the same operadic datum related by a level-dependent rescaling of the

Casimir, *not* by a rational-function identity in k: at $k = 0$ the trace form gives 0 while the KZ form gives $\Omega_{\mathrm{KZ}}/(h^\vee z) \neq 0$. The three parameter identifications below hold in the KZ convention; the trace-form analogues are obtained by the Sugawara rescaling $\hbar \mapsto \hbar \cdot 2h^\vee \cdot k(k + h^\vee)$.

1.1 The three parameters

Parameter 1: the KZ25 coupling. Khan and Zeng [1] study a 3d sigma model with holomorphic-topological boundary conditions. The boundary supports a Poisson vertex algebra $\mathrm{PVA}(\widehat{\mathfrak{g}}_k)$, and the classical r-matrix extracted from the λ-bracket is $r_{\mathrm{KZ}}(\lambda) = \Omega/(k + h^\vee)$. The coupling constant of the sigma model is $\hbar_{\mathrm{KZ}} = 1/(k + h^\vee)$.

Parameter 2: the DNP25 loop parameter. Dimofte, Niu, and Py [2] study line operators in 3d holomorphic-topological gauge theory. The meromorphic tensor product of two line operators V_{ℓ_1} and V_{ℓ_2} is governed by an R-matrix $R(z;\hbar)$ satisfying the quantum Yang–Baxter equation. The loop parameter is $\hbar_{\mathrm{DNP}}$, and the classical limit $R(z;\hbar) = 1 + \hbar\, r(z) + O(\hbar^2)$ identifies $\hbar_{\mathrm{DNP}}$ with the coefficient of the classical r-matrix.

Parameter 3: the collision-residue prefactor. The bar-cobar framework of [7] constructs the universal Maurer–Cartan element $\Theta_{\widehat{\mathfrak{g}}_k} := D_{\widehat{\mathfrak{g}}_k} - d_0$ from the bar differential. Its genus-zero binary collision residue $r(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta)$ is computed by extracting residues of the OPE against the bar propagator $d\log(z - w)$. The result is (1.1) with prefactor $\hbar_{\mathrm{bar}} = 1/(k + h^\vee)$.

1.2 Why the identification is non-trivial

The three parameters arise from different mathematical operations:

(i) $\hbar_{\mathrm{KZ}}$ from the λ-bracket of a Poisson vertex algebra (classical, no bar complex);

(ii) $\hbar_{\mathrm{DNP}}$ from the quantum Yang–Baxter equation of a line-operator R-matrix (quantum, categorical);

(iii) $\hbar_{\mathrm{bar}}$ from the residue of the bar differential against $d\log(z - w)$ (homological, operadic).

Their equality is not a priori obvious: (i) extracts a classical Poisson coefficient, (ii) a quantum deformation parameter, and (iii) a homological residue. That all three produce $1/(k + h^\vee)$ is a consequence of three independent theorems, one per link.

The collision residue $r(z) = \Omega/((k + h^\vee)z)$ lives on the ordered bar $B^{\mathrm{ord}}(\widehat{\mathfrak{g}}_k) = T^c(s^{-1}\bar{\widehat{\mathfrak{g}}}_k)$ with deconcatenation coproduct; it is an element of the E_1 convolution algebra. Parameter (iii) is extracted directly from this ordered datum, while parameter (ii) quantizes it via the Yangian R-matrix on the same ordered structure. The classical scalar shadow is

$$\kappa_{\mathrm{cl}} = \mathrm{av}(r(z)) = \frac{k\,\dim(\mathfrak{g})}{2h^\vee},$$

and the full modular characteristic is

$$\kappa(\widehat{\mathfrak{g}}_k) = \kappa_{\mathrm{cl}} + \frac{\dim(\mathfrak{g})}{2} = \frac{\dim(\mathfrak{g})(k + h^\vee)}{2h^\vee}.$$

2 The three extractions

2.1 Extraction 1: PVA lambda-bracket

The Poisson vertex algebra associated to $\widehat{\mathfrak{g}}_k$ has λ-bracket

$$\{J^a_\lambda J^b\} \;=\; f^{ab}{}_c\, J^c \;+\; k\,(t^a, t^b)\,\lambda, \tag{2.1}$$

where $\{J^a\}$ is a basis of $\mathfrak{g}$, $f^{ab}{}_c$ are the structure constants, and λ is the formal variable. The classical r-matrix is obtained by the Borel transform: the collision kernel of the λ-bracket at leading order in λ is

$$r_{\mathrm{KZ}}(z) \;=\; \frac{1}{k+h^\vee}\cdot\frac{\Omega}{z}. \tag{2.2}$$

The factor $1/(k+h^\vee)$ arises because the Sugawara construction normalizes the energy-momentum tensor by $(k+h^\vee)^{-1}$: the OPE of currents produces the Casimir tensor Ω at mode (1), and the λ-bracket (divided-power convention, $\{-_\lambda-\}_{(n)} = \lambda^n/n!$) yields the r-matrix with the Sugawara-normalized prefactor.

Proposition 2.1 (KZ25 coupling)**.** $\hbar_{\mathrm{KZ}} = 1/(k+h^\vee)$.

Proof. The sigma-model coupling in [1] is defined as the coefficient of Ω/z in the classical r-matrix of $\mathrm{PVA}(\widehat{\mathfrak{g}}_k)$. From (2.2), this is $1/(k+h^\vee)$. □

2.2 Extraction 2: line-operator R-matrix

The quantum R-matrix of [2] acts on the tensor product of line-operator representations:

$$R(z;\hbar) \;=\; 1 \;+\; \hbar\cdot\frac{\Omega}{z} \;+\; O(\hbar^2). \tag{2.3}$$

The QYBE $R_{12}(z_{12};\hbar)R_{13}(z_{13};\hbar)R_{23}(z_{23};\hbar) \;=\; R_{23}(z_{23};\hbar)R_{13}(z_{13};\hbar)R_{12}(z_{12};\hbar)$ holds at all orders in $\hbar$. At first order, it reduces to the classical Yang–Baxter equation for $r(z) = \hbar\,\Omega/z$. The loop parameter is the formal deformation parameter of the Yangian $Y_\hbar(\mathfrak{g})$.

Proposition 2.2 (DNP25 loop parameter)**.** $\hbar_{\mathrm{DNP}} = 1/(k+h^\vee)$.

Proof. The identification proceeds in two steps. First, [2] identifies the line-operator category at level k with the category of representations of the Yangian $Y_\hbar(\mathfrak{g})$ with $\hbar = \hbar_{\mathrm{DNP}}$. Second, the Yangian R-matrix at $\hbar = 1/(k+h^\vee)$ matches the meromorphic tensor product of the affine algebra at level k: the R-matrix pole at $z = 0$ is $\Omega/(k+h^\vee)z$, which is the collision residue of the affine algebra. The two conditions uniquely determine $\hbar_{\mathrm{DNP}} = 1/(k+h^\vee)$. □

2.3 Extraction 3: bar collision residue

The bar differential of $\widehat{\mathfrak{g}}_k$ extracts residues of the OPE against the bar propagator $d\log(z-w) = dz/(z-w)$. The OPE of two generating currents is

$$J^a(z)J^b(w) \;\sim\; \frac{k\,(t^a,t^b)}{(z-w)^2} \;+\; \frac{f^{ab}{}_c\,J^c(w)}{z-w}. \tag{2.4}$$

The bar propagator absorbs one pole, so the collision residue of the bar differential is

$$r_{\mathrm{bar}}(z) \;=\; \mathrm{Res}_{z=w}\Big[\frac{k\,(t^a,t^b)}{(z-w)^2}\cdot\frac{dz}{z-w}\Big]_{d\log} \;+\; (\text{simple-pole term}). \tag{2.5}$$

The $d\log$ extraction of the double-pole term gives $k\,(t^a,t^b)/z$ (one pole absorbed), and the simple-pole term contributes the structure-constant piece. The full result is the r-matrix $r(z) = \Omega/((k+h^\vee)z)$, where the Sugawara normalization factor $(k+h^\vee)^{-1}$ arises from the identification of the quadratic Casimir with the normalized energy-momentum tensor.

Proposition 2.3 (Bar collision prefactor)**.** $\hbar_{\mathrm{bar}} = 1/(k+h^\vee)$.

Proof. The collision residue $r(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta)$ is the genus-zero binary projection of $\Theta_{\widehat{\mathfrak{g}}_k} = D - d_0$. The linear part d_0 contributes the structure-constant terms $f^{ab}{}_c J^c/z$. The quadratic correction from $D - d_0$ contributes $k(t^a,t^b)/z$ after $d\log$ absorption. The normalization $\Omega = \sum_a t^a \otimes t_a$ with $(t^a, t_b) = \delta^a_b$ gives $r(z) = (k/(k+h^\vee))\cdot\Omega/z + (1/(k+h^\vee))\cdot$ [structure constants], which combines to $\Omega/((k+h^\vee)z)$ by the definition of Ω relative to the Killing form. The prefactor is $1/(k+h^\vee)$. □

3 The identification theorem

Theorem 3.1 (Three-parameter identification)**.** *For every simple Lie algebra $\mathfrak{g}$ and level $k \neq -h^\vee$, the three parameters*

$$\hbar_{\mathrm{KZ}}, \qquad \hbar_{\mathrm{DNP}}, \qquad \hbar_{\mathrm{bar}}$$

extracted from the Poisson vertex algebra coupling, the line-operator loop parameter, and the bar collision-residue prefactor, respectively, all equal

$$\hbar \;=\; \frac{1}{k+h^\vee}. \tag{3.1}$$

Proof. Propositions 2.1, 2.2, and 2.3. □

Remark 3.2 (Classical antecedents). The identification $\hbar = 1/(k+h^\vee)$ is not new for any single pair. The Sklyanin–Poisson bracket (Semenov-Tian-Shansky [4]) uses the Casimir tensor Ω with the level-shift normalization, and Drinfeld's quantization [3] introduces $\hbar$ as the Yangian deformation parameter satisfying $R(z;\hbar) = 1 + \hbar\,\Omega/z + O(\hbar^2)$. The classical limit $\hbar \to 0$ ($k \to \infty$) recovers the Poisson structure. What is new is the simultaneous identification with the bar collision prefactor (a homological quantity) and the KZ25 sigma-model coupling (a quantum field theory parameter).

Remark 3.3 (The critical level). At the critical level $k = -h^\vee$, all three parameters diverge: $\hbar \to \infty$. The collision residue develops a pole, the Sugawara construction becomes undefined, and the PVA λ-bracket degenerates. This is the Feigin–Frenkel critical-level phenomenon [5]: the center of the completed enveloping algebra at $k = -h^\vee$ is large (the Feigin–Frenkel center), and the classical r-matrix ceases to be well-defined. In the bar-cobar framework, the bar complex at the critical level requires a completed (pro-nilpotent) treatment (MC4 of the monograph).

4 Miura cross-universality: $(\Psi{-}1)/\Psi$ at all spins

The three-parameter identification of $\hbar$ on affine Kac–Moody has a companion identification on principal W-algebras that fixes the *cross-term coefficient* of the Drinfeld coproduct. This coefficient is the single datum relating $\hbar_{\text{Miura}}$ (the level-uniformizing variable on ${}_N$) to the three affine $\hbar$'s of Theorem 3.1.

Theorem 4.1 (Miura cross-universality)**.** *Let ${}_N$ be the principal W-algebra at level parameter*

$$\Psi \;=\; \frac{k+N}{k+N-1}, \qquad \Psi \neq 0, 1.$$

For every spin $s \geq 2$, the Drinfeld coproduct on the primary generator W_s contains the cross-term

$$\Delta_z(W_s) \;\supset\; \frac{\Psi-1}{\Psi}\bigl(J \otimes W_{s-1} \;+\; W_{s-1} \otimes J\bigr),$$

with the coefficient $(\Psi-1)/\Psi$ universal across all spins $s \geq 2$. *Composite corrections (normally-ordered products of lower-spin generators) appear at higher orders; the primary cross-term coefficient is s-independent and N-independent.*

Remark 4.2 (Coincidence-masking at $\Psi = 2$)*.* Earlier drafts wrote the cross-term as $(1/\Psi)\bigl(J \otimes W_{s-1} + W_{s-1} \otimes J\bigr)$. At $\Psi = 2$ this coincides numerically with $(\Psi-1)/\Psi = 1/2$, the coincidence that masked the error. The Beilinson discipline substitutes $\Psi = 3$: then $(\Psi-1)/\Psi = 2/3$ but $1/\Psi = 1/3$. A direct Miura computation at $N = 3$, spin $s = 3$, $\Psi = 3$ (verifiable by elementary symmetric expansion of $T(u) = \prod_{i=1}^{3}(u + \Lambda_i)$) reads off $2/3$, not $1/3$.

Remark 4.3 ($\Psi = 0, 1$ degenerations)*.* The two special values of Ψ sit at the endpoints of the Miura quantization line:

(i) $\Psi = 1$ (free boson): $(\Psi-1)/\Psi = 0$, the cross-term vanishes, and Δ_z collapses to the plain coproduct on an abelian Heisenberg parent. The OPE obstruction to chiral quantization disappears.

(ii) $\Psi = 0$ ($k = -N$, free-abelian degeneration): the coefficient diverges as a rational function of Ψ; the chiral algebra degenerates to an abelian Heisenberg factor with identity quantum group, and the Miura coefficient is not defined.

Proposition 4.4 ($\hbar_{\text{Miura}}$ as the fourth parameter)**.** *The* Miura coupling

$$\hbar_{\text{Miura}} \;:=\; \frac{\Psi-1}{\Psi} \;=\; \frac{1}{k+N}$$

is the principal-W companion of the three affine couplings. At $\mathfrak{g} = \mathfrak{sl}_N$ and k non-critical, $\hbar_{\text{Miura}} = \hbar_{\text{KZ}}/(k+N-h^\vee+h^\vee) = 1/(k+N) = \hbar_{\text{KZ}}$, so the four couplings all reduce to the shifted-level reciprocal $1/(k+h^\vee)$ on $\mathfrak{sl}_N$ (where $h^\vee = N$).

Proof. Direct: $\Psi - 1 = 1/(k+N-1)$ inverted gives $\Psi = (k+N)/(k+N-1)$, hence $(\Psi-1)/\Psi = 1/(k+N)$. For $\mathfrak{sl}_N$, $h^\vee = N$, so $k+N = k+h^\vee$ and the Miura coupling equals $\hbar_{\text{KZ}}$. □

5 Four Yangian types must be kept distinct

The three parameters of Theorem 3.1 extract the *same* coupling from three *different* algebraic structures. A prerequisite for stating this precisely is distinguishing four Yangian notions that the programme uses interchangeably in less-disciplined expositions.

Definition 5.1 (Four Yangian types)**.** We fix the following four types with superscript discipline:

(1) $Y_\hbar(\mathfrak{g})$ *classical*: the E_1-topological Yangian on , Drinfeld's 1985 first presentation.

(2) $Y_\hbar^{\mathrm{dg}}(\mathfrak{g})$ *dg-shifted*: the derived Yangian at a point or on the formal disk, carrying a shifted symplectic structure.

(3) $Y_\hbar(\mathfrak{g})^{\mathrm{ch}}$ *chiral*: E_1-chiral on a smooth curve X, presented as a D-module with spectral parameter in the ordered collision coordinate. This is the home of the Yangian for the Drinfeld–Kohno bridge.

(4) $Y_\hbar(\mathfrak{g})^{\mathrm{spec}}$ *spectral*: the factorization algebra on $\mathbb{A}^1_u$ obtained by transposing the spectral coordinate with the worldsheet coordinate.

The three-parameter identification of Theorem 3.1 is between $\hbar$-parameters of type (3), the chiral Yangian $Y_\hbar(\mathfrak{g})^{\mathrm{ch}}$.

Remark 5.2 (Two Yangian definitions on the chiral type). Within type (3), two abstract definitions coexist: (a) an E_1-factorization algebra on X together with an RTT presentation, and (b) a quadruple $(\mathcal{A}, S(z), \Delta^{\mathrm{ch}}, \{m_k\})$ consisting of a chiral algebra, an S-operator, a chiral coproduct, and a modular MC tower. Definition (b) is strictly stronger because the MC tower is not derivable from the RTT data alone. The three $\hbar$'s of Theorem 3.1 agree under both definitions: (a) sees $\hbar$ as the RTT deformation parameter; (b) sees it as the leading coefficient of m_2. Equality of the two definitions on the ordered Koszul locus is the content of the chiral QG equivalence theorem.

6 Verification

Computation 6.1 (Explicit checks)**.** The three-parameter identification is verified numerically for the following families:

$\mathfrak{g}$	$h^\vee$	k (test values)	$\hbar = 1/(k+h^\vee)$
$\mathfrak{sl}_2$	2	$1, 2, 5, 10, -1/2$	$1/3, 1/4, 1/7, 1/12, 2/3$
$\mathfrak{sl}_3$	3	$1, 2, 5$	$1/4, 1/5, 1/8$
$\mathfrak{sl}_4$	4	$1, 3$	$1/5, 1/7$
$\mathfrak{so}_5$	3	$1, 2$	$1/4, 1/5$
G_2	4	1	$1/5$
E_8	30	1	$1/31$

For each entry, the compute layer verifies:

(i) the PVA λ-bracket coefficient matches $\Omega/(k+h^\vee)$;

(ii) the Yangian R-matrix classical limit matches $\Omega/(k+h^\vee)z$;

(iii) the bar collision residue matches $\Omega/(k+h^\vee)z$.

Total: 39 tests across 6 Lie algebras and 13 level values.

Remark 6.2 (Convention dependence). The formula $\hbar = 1/(k+h^\vee)$ depends on the normalization of the Killing form. We use the convention where long roots have squared length 2, which gives $h^\vee(\mathfrak{sl}_N) = N$. Other conventions (e.g., $\mathrm{tr}_{\mathrm{fund}} = 1/2$) rescale k and $h^\vee$ simultaneously, leaving $\hbar$ invariant. The divided-power convention ($\lambda^{(n)} = \lambda^n/n!$ in the λ-bracket) is absorbed into the definition of r_{KZ}; the OPE-mode convention gives $J^a_{(1)}J^b = k(t^a, t^b)$ (no factorial).

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